MEA OCPP 1.6 Compliance Test Report

Section 7: Abnormal Operation Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

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1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Plug Timeout	0 s per plug event
Boot Wait Timeout	90 s (after Hard Reset)
Local Stop Timeout	30 s
Test Date	2026-06-10 08:20:29 UTC
Results file	tex/vsecc_section7_results.json

Test Architecture

The test runs directly against the MEA CSMS with no proxy. The test PC:

- Calls the vSECC REST API (`http://192.168.1.166/api`) for login and state queries
- Subscribes to the vSECC MQTT broker (`mqtt://192.168.1.166:1883`) to observe StatusNotification, charging session state, and meter values
- Calls the MEA sandbox REST API (`https://ocppapi.measandbox.com/EV/`) with HTTP Digest authentication for RemoteStart, RemoteStop, Reset, and SendLocalList

Several sub-scenarios (7.3, 7.4, 7.5) require physical hardware interaction (RFID card scan, emergency stop button, door open switch) that cannot be triggered programmatically. Those items are recorded as **WARN** with a remark describing the required physical action. Sub-scenario 7.6 simulates power loss by issuing a Hard Reset via the MEA REST API.

2 Section 7 Results

Summary: 8 PASS 1 FAIL 45 WARN 0 SKIP (54 total)

Item	Test Description	Detail / Evidence	Result
7.1 — RemoteStart while Unplugged (Available State)			
7.1.1	StatusNotification Available (charger idle, no EV)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	**WARN**
7.1.2	RemoteStartTransaction (EV not connected) → HTTP 200	`HTTP 200` *vSECC should respond Rejected when no EV present*	**PASS**
7.2 — Concurrent RemoteStart (Second Rejected)			
7.2.1	StatusNotification Available (before EV plug)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	**WARN**
7.2.2	StatusNotification Preparing (EV plug)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	**WARN**
7.2.3	RemoteStartTransaction (first) → HTTP 200	`HTTP 200`	**PASS**
7.2.4	RemoteStartTransaction (second, concurrent) → HTTP 200	`HTTP 200` *CSMS likely blocks duplicate; charger-level Rejected not observable without pro...*	**PASS**
7.2.5	StartTransaction (session started after first RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	**WARN**

Item	Test Description	Detail / Evidence	Result
7.2.6	StatusNotification Charging	`Charging status not observed on MQTT in 20 s`	WARN
7.2.7	MeterValues (during charging)	`MeterValues not observed on MQTT in 15 s`	WARN
7.2.8	RemoteStopTransaction → HTTP 200	`HTTP 200`	PASS
7.2.9	StopTransaction (session ended after RemoteStop)	`session_state idle/finished not observed on MQTT in 25 s`	WARN
7.2.10	StatusNotification Finishing	`Finishing not observed on MQTT in 15 s` *Some implementations skip Finishing and go directly to Available*	WARN
7.2.11	StatusNotification Available (after stop / unplug)	`Available not observed on MQTT in 30 s | unplug EV`	WARN
7.3 — Swap Card (Invalid then Valid RFID)			
7.3.1	StatusNotification Available (before EV plug)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	WARN
7.3.2	StatusNotification Preparing (EV plug)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
7.3.3	Authorize (invalid RFID tag) → Rejected	`Not triggerable via API | vSECC reads RFID directly from hardware` *Requires physical RFID scan on vSECC hardware of unknown/blocked card*	WARN
7.3.4	Authorize (valid RFID tag) → Accepted	`Not triggerable via API | vSECC reads RFID directly from hardware` *Requires physical RFID scan on vSECC hardware of authorized card after rejected...*	WARN
7.3.5	StartTransaction (after valid RFID)	`Depends on 7.3.3/7.3.4 (physical RFID scan)` *Session starts only after valid RFID tap on vSECC hardware*	WARN
7.3.6	StatusNotification Charging (after valid auth)	`Depends on 7.3.5 (StartTransaction)` *Requires completed Authorize sequence via physical RFID hardware*	WARN
7.3.7	MeterValues (during charging)	`Depends on 7.3.6 (Charging state)` *Requires completed Authorize sequence via physical RFID hardware*	WARN
7.3.8	RemoteStopTransaction	`Depends on 7.3.5 (StartTransaction)` *Requires active session started via physical RFID hardware*	WARN
7.3.9	StopTransaction (after RemoteStop)	`Depends on 7.3.8 (RemoteStop)` *Requires active session started via physical RFID hardware*	WARN
7.3.10	StatusNotification Finishing	`Depends on 7.3.8 (RemoteStop)` *Requires active session started via physical RFID hardware*	WARN
7.3.11	StatusNotification Available (after stop / unplug)	`Depends on 7.3.8 (RemoteStop)` *Requires active session started via physical RFID hardware*	WARN
7.4 — Emergency Stop			
7.4.1	StatusNotification Available (before EV plug)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	WARN

Item	Test Description	Detail / Evidence	Result
7.4.2	StatusNotification Preparing (EV plug)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
7.4.3	RemoteStartTransaction → HTTP 200	`HTTP 200`	PASS
7.4.4	StartTransaction (session started after RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	WARN
7.4.5	StatusNotification Charging	`Charging status not observed on MQTT in 20 s`	WARN
7.4.6	MeterValues (during charging)	`MeterValues not observed on MQTT in 15 s`	WARN
7.4.7	StopTransaction (reason=EmergencyStop)	`Triggered by physical emergency stop button on vSECC hardware` *Requires physical emergency stop button press on the vSECC unit*	WARN
7.4.8	StatusNotification Faulted (after EmergencyStop)	`Faulted status not observed on MQTT (E-stop not pressed)` *Requires physical emergency stop button press on the vSECC unit*	WARN
7.4.9	StatusNotification Available (E-stop cleared)	`Depends on 7.4.7 (physical E-stop press and release)` *E-stop must be physically released to restore normal operation*	WARN
7.4.10	StatusNotification Available (EV unplug after E-stop recovery)	`Depends on 7.4.7 (physical E-stop press and release)` *EV must be unplugged after E-stop recovery to reach Available*	WARN
7.5 — Open Door			
7.5.1	StatusNotification Available (before door open)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	WARN
7.5.2	StatusNotification Faulted (reason=DoorOpen)	`Triggered by opening physical door/panel on vSECC hardware` *Requires physical door open on vSECC hardware; not triggerable via API*	WARN
7.5.3	StatusNotification Available (after door closed)	`Depends on 7.5.2 (physical door open and close)` *Door must be physically closed to restore normal operation*	WARN
7.6 — Power Loss / Reboot (Simulated via Hard Reset)			
7.6.1	StatusNotification Available (before EV plug)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	WARN
7.6.2	StatusNotification Preparing (EV plug)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
7.6.3	RemoteStartTransaction → HTTP 200	`HTTP 200`	PASS
7.6.4	StartTransaction (session started after RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	WARN
7.6.5	StatusNotification Charging	`Charging status not observed on MQTT in 20 s`	WARN
7.6.6	MeterValues (during charging)	`MeterValues not observed on MQTT in 15 s`	WARN

Item	Test Description	Detail / Evidence	Result
7.6.7	StopTransaction (PowerLoss) — simulated via Hard Reset → HTTP 200	`HTTP 200` *Simulated via Hard Reset; physical power loss not testable in this environment*	PASS
7.6.8	BootNotification → Accepted (vSECC reconnected after reboot)	`CSMS reconnection confirmed after Hard Reset`	PASS
7.6.9	StatusNotification Preparing (EV still connected after reboot)	`Preparing not observed on MQTT in 20 s after reboot` *EV must remain connected through the reboot sequence*	WARN
7.6.10	StatusNotification Available (EV unplug after reboot)	`Available not observed on MQTT in 30 s | unplug EV`	WARN
7.7 — Local List Offline			
7.7.1	SendLocalList (version=1, RFID_TEST=Accepted) → HTTP 200	`HTTP 404`	FAIL
7.7.2	StatusNotification Available (charger online before offline test)	`Offline sequence not observable via MQTT/REST without proxy` *Offline buffering not observable via MQTT/REST without proxy*	WARN
7.7.3	StatusNotification Preparing (EV plug while charger offline)	`Offline start requires disconnecting vSECC from CSMS network` *Offline buffering not observable via MQTT/REST without proxy*	WARN
7.7.4	Authorize (local list lookup while offline) → Accepted	`vSECC must authorize from local list while CSMS unreachable` *Offline buffering not observable via MQTT/REST without proxy*	WARN
7.7.5	StartTransaction (offline, buffered by vSECC)	`vSECC buffers transaction in firmware; not observable without proxy` *Offline buffering not observable via MQTT/REST without proxy*	WARN
7.7.6	StopTransaction (offline, buffered by vSECC)	`vSECC buffers stop in firmware; delivered when CSMS reconnects` *Offline buffering not observable via MQTT/REST without proxy*	WARN
7.7.7	CSMS reconnect — buffered Start/StopTransaction flushed	`Reconnect flushing is firmware-internal; not observable without proxy` *Offline buffering not observable via MQTT/REST without proxy*	WARN

3 Notes on Failures and Warnings

7.1 — RemoteStart while unplugged

- **7.1.2** The MEA REST API call returns HTTP 200 if the command was delivered to the vSECC. The vSECC OCPP layer should respond `Rejected` to the CSMS because no EV is present. This rejection is observable only via a proxy intercepting the OCPP WebSocket; therefore the item is recorded as PASS for HTTP 200 delivery with a WARN remark about the expected Rejected response.

7.2 — Concurrent RemoteStart

- **7.2.4** A second RemoteStart while a session is already active will be blocked by the CSMS before reaching the charger. Whether the charger itself responds `Rejected` is not observable without a proxy. The item is recorded as WARN.

- **7.2.5–7.2.11** Depend on the first RemoteStart being accepted and the EV being physically connected (item 7.2.2).

7.3 — Swap Card

- **7.3.3, 7.3.4** The vSECC reads RFID cards directly via its hardware reader. There is no REST or MQTT API to inject an RFID scan. Both items (invalid card, then valid card) require the operator to physically tap RFID cards on the vSECC unit.
- **7.3.5–7.3.11** All subsequent items in this sub-scenario depend on the completed RFID swap sequence and are therefore WARN.

7.4 — Emergency Stop

- **7.4.7** The emergency stop is a physical button on the vSECC hardware. There is no software API to trigger it. Items 7.4.8–7.4.10 depend on the E-stop being pressed and released physically and are recorded as WARN.

7.5 — Open Door

- **7.5.2** The door-open fault is triggered by a physical door/panel switch on the vSECC unit. There is no software API to simulate this condition. Item 7.5.3 depends on the door being physically closed and is WARN.

7.6 — Power Loss / Reboot

- **7.6.7** Physical power loss cannot be reproduced in a controlled automated test without additional hardware. A Hard Reset via `POST /EV/remote/reset` is used as the closest available approximation. The item is recorded as PASS if the MEA REST API returns HTTP 200, with a WARN remark.
- **7.6.8** After the Hard Reset the vSECC reboots. The test waits up to `BOOT_WAIT_SEC` seconds for reconnection, confirmed by the MQTT `vsecc/ocpp_connection_status=connected` event or a successful REST API probe.
- **7.6.9, 7.6.10** EV must remain plugged in through the reboot and then be unplugged to trigger the final Available transition.

7.7 — Local List Offline

- **7.7.1** SendLocalList is issued via `POST /EV/remote/sendLocalList` on the MEA REST API. HTTP 200 confirms the command was accepted by the CSMS and relayed to the vSECC.
- **7.7.2–7.7.7** Offline start, local RFID authorization, transaction buffering, and reconnect flushing are handled entirely within the vSECC firmware. Without a proxy intercepting the OCPP WebSocket it is not possible to observe these behaviors from the test PC. All items are recorded as WARN.

4 Dependency Map

7.1 Independent (no EV required)

7.2 7.2.2 (EV plug) → 7.2.3 (RemoteStart 1) → 7.2.5 → 7.2.6 → 7.2.7 → 7.2.8 (RemoteStop) → 7.2.9 → 7.2.10 → 7.2.11

7.3 7.3.2 (EV plug) → 7.3.3/7.3.4 (physical RFID) → 7.3.5 → . . . → 7.3.11

7.4 7.4.2 (EV plug) → 7.4.3 (RemoteStart) → 7.4.4 → 7.4.5 → 7.4.6 → 7.4.7 (physical E-stop) → 7.4.8 → 7.4.9 → 7.4.10

7.5 7.5.2 (physical door open) → 7.5.3

7.6 7.6.2 (EV plug) → 7.6.3 (RemoteStart) → 7.6.4 → 7.6.5 → 7.6.6 → 7.6.7 (Hard Reset) → 7.6.8 (BootNotification) → 7.6.9 → 7.6.10

7.7 7.7.1 (SendLocalList) independent; 7.7.2–7.7.7 require offline network isolation